

SatStack

App Proposal — Idea & Functionality

App Idea

SatStack is a privacy-first Bitcoin DCA (Dollar-Cost Averaging) journal and analytics app designed to help Bitcoiners track their stack and evaluate market sentiment.

Problem Statement

Most crypto portfolio apps rely on centralised cloud APIs that compromise operational security, lack physical privacy features, and clutter the UI with altcoins. SatStack solves this by providing a strictly **offline-first, locally stored** DCA journal secured by biometrics, paired with a clean market sentiment indicator to help users time their purchases safely.

Core Functionalities

Onboarding (First Launch)

A 4-page `HorizontalPager` shown only on first launch. Guides the user through selecting their fiat currency, setting a milestone goal (with optional sample data), and entering an exchange URL. All selections are persisted to DataStore; subsequent launches skip directly to the Vault.

The Vault (Launch Screen)

App entry is secured by native device biometrics via `AndroidX BiometricPrompt`.

Tab 1 — Home (Dashboard & History)

- Displays total accumulated Satoshis
- Visual progress bar tracking completion percentage towards the user's Milestone Goal (set in Settings)
- Scrollable list of past manual DCA entries with swipe-to-delete

Floating Action Button (FAB) opens the **Add DCA Entry** form as a **modal bottom sheet** — slides up over the Dashboard without navigating away, dismissed by saving, swiping down, or tapping outside.

Tab 2 — Analytics

- Fetches and displays the live Fear & Greed Index via the alternative.me API

- Helps users gauge market sentiment before buying
- Portfolio card — displays live BTC price (CoinGecko API), stack value, total spent, and P&L (colour-coded green/red)
- "Buy More Sats" button opens the user's saved exchange URL in the browser

Tab 3 — Settings

- Set personal Stacking Milestone Goal (e.g. 1,000,000 sats)
- Save preferred Exchange URL (e.g. Strike, Swan)
- Select preferred fiat currency (GBP / USD)
- Toggle Dark / Light mode

Technical Requirements Mapping

Functionality	Requirement Met
Vault, 3-Tab Bottom Nav (Home, Analytics, Settings), Add Data screen	Minimum two distinct screens
FAB → Add DCA Entry bottom sheet, Bottom Nav routing	Navigation component
"Buy More" button launching the Exchange URL saved in Settings	Intent to external browser
Preserving form inputs & display states on rotation	App lifecycle / screen rotation
Biometric lock on launch screen	Permissions used responsibly / Mobile App Security
Internet permission for Analytics API	Permissions used responsibly
Saving transactions (Date, Fiat, Sats) to Room/SQLite DB	Local storage
DB read/write/delete operations for Home Tab history	Local storage
Shared access to transaction data	ContentProvider + instrumented tests

Optional Features (3 Selected)

Touch Gestures

Double-tap on the total balance (Home Tab) triggers **Privacy Mode** — instantly masks the balance with *** and updates a visual privacy-eye icon.

Notifications

The app checks the user's total balance against the Milestone Goal defined in Settings. When the threshold is crossed, it fires a local notification reminding the user to practice self-custody.

Broadcast Events

Listens for network state changes. If offline, the Analytics tab displays an **"Offline Mode"** banner while the Home Tab (local DB) remains fully functional.

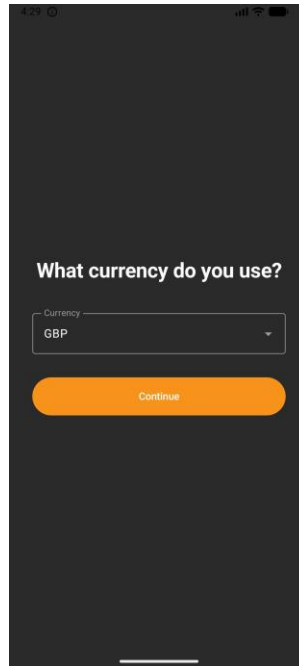
App Design

Onboarding Flow

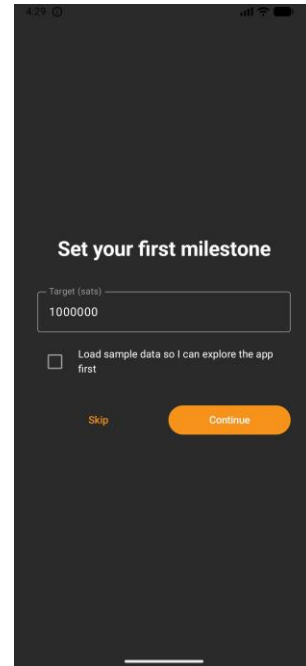
4-page first-launch wizard shown only once, guiding the user through currency selection, stacking milestone, and exchange URL setup.



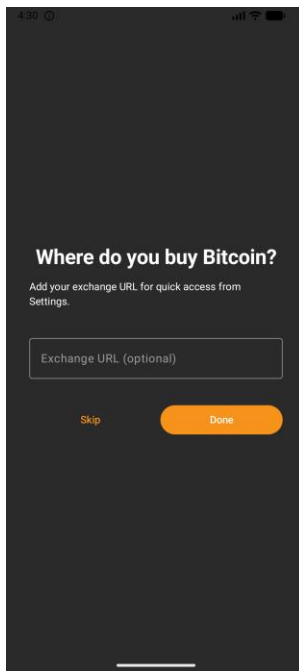
Welcome Screen



Currency Selection



Milestone Goal



Exchange URL

The Vault — Launch Screen

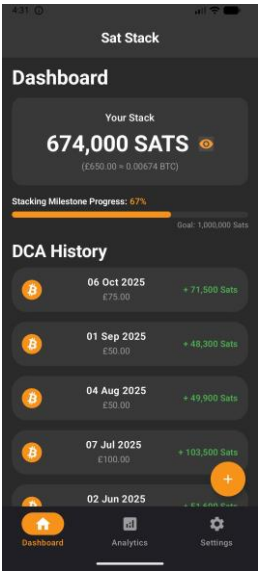
Every app launch is gated by native biometric authentication via AndroidX BiometricPrompt, ensuring the journal remains private even if the device is accessed by others.



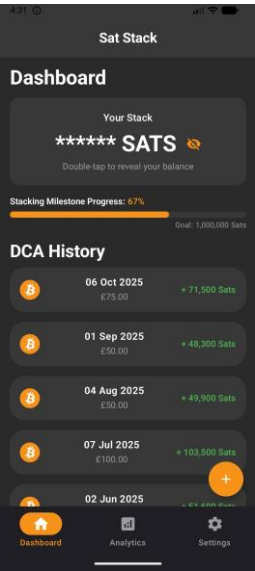
Biometric Lock Screen

Tab 1 — Dashboard & Add DCA Entry

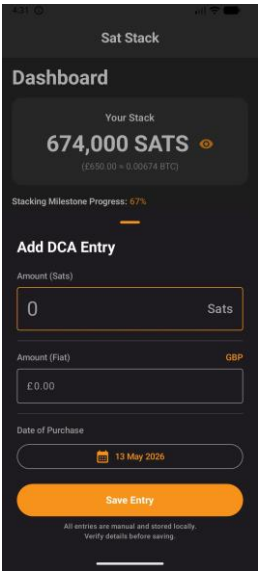
Home screen displaying total Satoshis, milestone progress, and a scrollable DCA history. The FAB slides up the Add DCA Entry bottom sheet without leaving the Dashboard. Double-tapping the balance activates Privacy Mode.



Dashboard (Balance Visible)



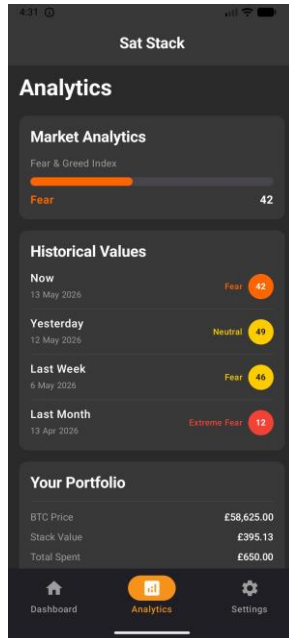
Privacy Mode Active



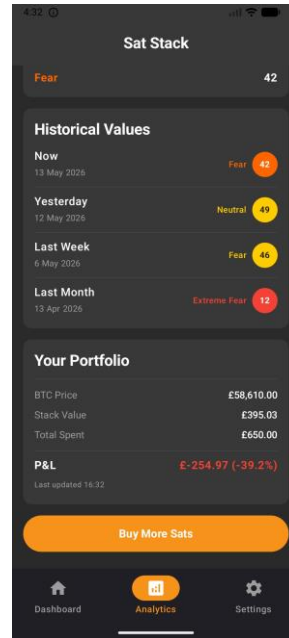
Add DCA Entry Form

Tab 2 — Analytics

Live Fear & Greed Index with historical sentiment values, a portfolio card showing BTC price, stack value, total spent and colour-coded P&L, plus a Buy More Sats shortcut to the user's saved exchange.



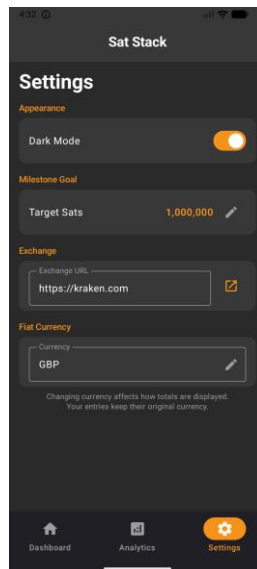
Market Analytics & Historical Values



Portfolio Card & Buy More Sats

Tab 3 — Settings

Persistent user preferences: dark/light mode toggle, stacking milestone target, exchange URL, and fiat currency selection. All settings are stored locally via DataStore.



Settings Screen

